Based on 49 sources

Concept	Description	Formula or Identity
Information Theory (Shannon)	The foundation of communication systems, defined by the study of message transmission, entropy, and channel capacity in discrete and continuous systems.	$H = -\sum p_i \log p_i$
Entropy	A measure of the uncertainty or information content associated with a discrete random variable.	$H(X) = -\sum p(x) \log p(x)$
Channel Capacity	The maximum rate at which information can be transmitted over a communication channel with an arbitrarily small error probability.	$C = B \log_2(1 + S/N)$
Equivocation	The conditional entropy of the transmitted message	$H_y(x) = H(x, y) - H(y)$